INCH-POUND

MIL-DTL-22529/1C
6 February 1998
SUPERSEDING
MIL-DTL-22529/1B
23 September 1997

DETAIL SPECIFICATION SHEET

GROMMET, COMPOSITE, EDGING

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-22529 listed in that issue of the Department of Defense Index of Specification and Standards (DoDISS) specified in the solicitation.

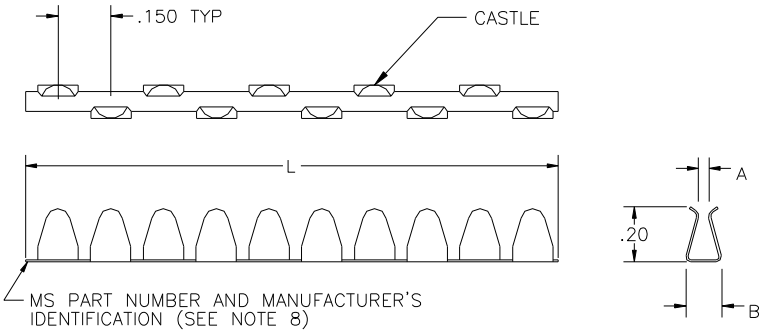


FIG. 1

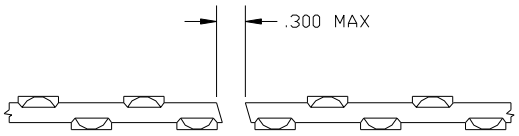
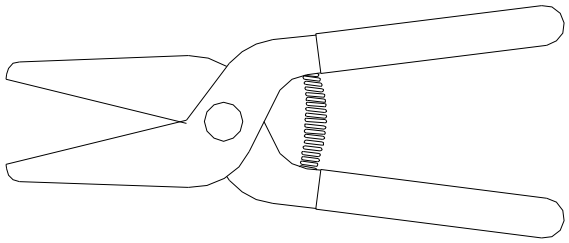


FIG. 2



SHEARS, FULL BYPASS
MFE-100

FIG. 3

Beneficial comments, recommendations, additions, deletions, clarifications, etc., And data which may improve this document should be sent to: Defense Industrial Supply Center, attn.: DISC-ECC, 700 Robbins Avenue, Philadelphia, PA 19111-5096.

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Table 1
Military Specification
Part number & Cross reference

Military Standard		Sheet	Unit		Device Technologies, INC. Marborough, MA 01752 Cage Code O-HES-O	
Part Number	Dash Variation	Thickness A	Length L	B	Part Number	
M22529/1	-1R-25	.025	25 Ft.	.14	T1-0025	-ER06
	-1R-100	.036	100 Ft.		T1-0100	
	-2R-25	.036	25 Ft.		T2-0025	
	-2R-100	.063	100 Ft.		T2-0100	
	-3R-25	.059	25 Ft.		T3-0025	
	-3R-100	.074	100 Ft.		T3-0100	
	-4R-25	.070	25 Ft.	.19	T4-0025	
	-4R-100	.093	100 Ft.		T4-0100	
	-5R-25	.090	25 Ft.		T5-0025	
	-5R-100	.111	100 Ft.		T5-0100	
	-6R-25	.105	25 Ft.		T6-0025	
	-6R-100	.134	100 Ft.		T6-0100	
	-7R-25	.178	25 Ft.	.25	T7-0025	
	-7R-100	.198	100 Ft.		T7-0100	
	-8R-25	.240	25 Ft.	.32	T8-0025	
	-8R-100	.260	100 Ft.		T8-0100	
	-nC-85	--	12.75 In.	--	Tn-0085	-EC06
MFE-100		Shears full bypass			#830	

Table 2
Circular hole
grommet edging cut lengths

Note: cut length is a multiple of .15 inches (Fig 1)

Nominal Hole Diameter Reference	Cut Length L Length	Cut Length ln Castles
2.000	6.00	40
2.250	6.90	46
2.500	7.65	51
2.750	8.40	56
3.000	9.15	61
3.250	10.05	67
3.500	10.80	72
3.750	11.55	77
4.000	12.30	82
4.250	13.20	88
4.500	13.95	93
4.750	14.70	98
5.000	14.45	103
5.250	16.35	109
5.500	17.10	114
5.750	17.85	119
6.000	18.60	124

Requirements:

1 Materials:

1.1 Substrate: -1 thru –7, ¼ hard 300 series stainless steel per AMS 5517, AMS 5903, AMS 5907 or AMS 5310; -8, ½ hard 300 series stainless steel per AMS 5518, AMS 5904 or AMS 5911.

1.2 Coating: Thermosetting epoxy, color lime green per FED-STD 595, Color Number 14533.

1.2.1 Specific Gravity (ASTM -D-792) Powder 1.21.

1.2.2 Impact Resistance (ASTM A775) .125 x 3 in. x 3 in. 160 in. lbs.

1.2.3 Penetration (ASTM G17) -40° to 240°F Zero.

1.2.4 Abrasion Resistance (ASTM 01044) wheel type CS-17, 1000g weight, 5000 cycles - .008g loss

1.2.5 Elongation: No cracks @ 15% Elongation.

1.2.6 Thermal Endurance: Per (IEEE 57) 145 °C (20,000 Hrs)

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1.2.7 Outgassing

Total Mass Loss	.03%
Vapor Condensed Molecule	.00%

1.2.8 Salt spray resistance per MIL-STD-202, method 101D.6.3-500 hrs.

2. The Composite Grommet meets all the requirements of MIL-DTL-22529 (AS) with the following exception: 4.2.9 Dielectric withstanding voltage, 750 Volts, 60 Hz. This grommet is primarily intended to provide chafing protection.

3. Examples of Part Number:

3.1 Reels:

M22529/1 -4 R –25 (T4-0025-ER06)

- -4: Sheet thickness .070”-.093”
- R: Reel
- –25: Length in feet

3.2 Cut Lengths:

Cut Lengths are available in increments of .15”. EXAMPLE: Calculated Length: 4.10” dia. Hole x π = 12.88”/.15 – 85.87 castles, cut shorter to the nearest whole castle.

M22529/1 -2 C -85 (T2-0085-EC06)

- -2: Sheet thickness .036”-.063”
- C: Factory Precut Length
- -85: Number of whole castles 85 x .15 = 12.75”

Notes:

1 Tolerances: .XX = ±.02; .XXX = ±.010

2 The Government shall procure stock grommet edging on 25’ and 100’ reels or in Factory pre-cut lengths per paragraph 3.2 cut length 12.75”.

3 The installing activity shall fabricate grommets using MFE-100 shears per Fig 3.

4 Within a given sheet thickness range and spring characteristic of the grommet, end user should determine the grommet size by trial.

5 Composite Grommet Edgings are to be used for edge dressing, odd-shaped holes, and circular holes to provide chafing protection.

6 Installation:

6.1 The maximum allowable end gap for odd-shaped and circular holes after grommet edging is installed flush against the hole periphery is per Fig 2.

6.2 When the gap must be closed, overlap and interlock the terminal ends of the grommet.

6.3 Installation instructions are to be included with each package.

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6.4 When installing grommets, the gap (joint) shall be at the top of the cutout hole. Avoid locating joints in the lower half of the cutout hole.

7. U.S. Patent No's 4,769,966 and 5,343,669 expire 20 April 2007 and 6 July 2012 respectively. Other U.S. and foreign patents pending. The Government does not have a royalty-free license.

8 The military detail part number, single-digit dash variation, and manufacturer's identification to be printed and repeated on the surface of the grommet at least every 10.6 in. per Fig 1.

9 Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last issue.

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